



LLM-derived metrics in second language writing assessment: an explainable AI approach

Jingying Hu^a , Yan Cong^{a,b,*} 

^a Department of Linguistics, Purdue University, USA

^b School of Languages and Cultures, Purdue University, USA

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ABSTRACT

Large language models (LLMs) are increasingly used in educational technology for automated writing assessment, yet most applications rely on prompt-based scoring and feedback generation, which often lack transparency, reproducibility, and interpretability. This study investigates whether model-internal LLM representations can provide interpretable and reproducible metrics for second language (L2) writing assessment. We derived surprisal and perplexity from next-token prediction to quantify linguistic predictability and embedding-based similarity to measure semantic coherence across sentences. These metrics were computed at the token, sentence, and discourse levels using three pretrained Chinese language models and evaluated on 1196 essays written by Chinese L2 learners across four proficiency levels. Their relationships with 11 established linguistic measures of fluency, lexical sophistication, phraseological complexity, and syntactic complexity were also examined. Results showed that surprisal and perplexity generally decreased with proficiency for the two Traditional Chinese-focused models, indicating greater linguistic predictability in more proficient writing, whereas the multilingual model showed weaker sensitivity. Embedding-based similarity increased with proficiency, reflecting stronger semantic coherence. Combining LLM-derived metrics with classical linguistic features improved proficiency classification and prediction beyond either feature set alone. Correlation and qualitative analyses further demonstrated that the proposed metrics capture complementary aspects of writing while revealing conditions under which their interpretations become less reliable. These findings demonstrate the value of interpretable, model-derived metrics for transparent, reproducible, and scalable AI-supported L2 writing assessment, particularly for underrepresented learner populations and lower-resource languages.

1. Introduction

Artificial intelligence has become increasingly integrated across learning, teaching, assessment, and educational administration (Chiu et al., 2023). Automated writing assessment, in particular, offering scalable ways to characterize learner performance and support scoring and feedback. In second language (L2) writing research, automated assessment has traditionally relied on a large set of well-established linguistic features capturing complexity, accuracy, fluency, and coherence to describe developmental patterns or

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* Corresponding author. School of Languages and Cultures, Purdue University, West Lafayette, IN, 47907, USA.

E-mail address: cong4@purdue.edu (Y. Cong).

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evaluate learner performance (Crossley et al., 2016; Hao et al., 2024; Kyle & Crossley, 2015, 2018). Although these features are useful, each typically captures an isolated or a limited aspect of writing and may not transfer robustly across languages. Recent advances in artificial intelligence, particularly large language models (LLMs), have expanded automated writing assessment to prompt engineering-based essay scoring, grammatical error detection, and personalized feedback generation (Deng et al., 2025; Li & Liu, 2024; Qiu et al., 2024; Saricaoglu & Bilki, 2025; Stahl et al., 2024). However, prompting-based approaches often suffer from limited transparency and stability, as prompted outputs can vary with prompt design and the process underlying model judgement remains opaque (Stahl et al., 2024; Tang et al., 2024). From an educational technology perspective, the central issue is therefore not merely whether LLMs can reproduce human-assigned scores, but whether the assessment evidence they generate is sufficiently transparent, stable, and reproducible to support defensible decisions. These requirements are particularly important in high-stakes educational contexts, where fairness and accountability are essential.

Within the explainable artificial intelligence (XAI) (Arrieta et al., 2020; Calderon & Reichart, 2025) framework, recent studies have therefore begun moving beyond prompt-based scoring by probing model-internal mechanisms of LLMs to reveal and unpack their black-box behavior. These approaches aim to derive interpretable, quantitative metrics directly from models' internal representation and interpretation of language. Specifically, two types of metrics are particularly promising and relevant to writing assessment: (1) probabilistic measures such as surprisal and perplexity quantify how predictable a linguistic unit is under a language mode, therefore they provide hypothetically holistic indices of writing naturalness by reflecting the extent to which language use conforms to distributional patterns learned from large-scale native-speaker training data (Cong, 2025; Hu & Cong, 2025; Sánchez et al., 2024); (2) embedding-based similarity, operationalized as cosine similarity between LLM-derived contextualized sentence representations, quantifies gradient semantic relatedness of writing and can potentially serve as a proxy for discourse coherence (Cong, 2024).

The present study extends prior work, which has focused primarily on English, to L2 Chinese writing assessment. As shown in Fig. 1, we operationalized LLM-derived surprisal, perplexity and similarity metrics across token-, sentence-, and discourse-level and validated their effectiveness tracking L2 writing development for (traditional) Chinese. We further tested whether these measures add predictive information beyond classical linguistic metrics through controlled ablation analyses. Finally, we conducted correlational analysis and qualitative error analysis of representative learner essays to illustrate how the proposed metrics relate to and characterize constructs in L2 learning.

The present study has important methodological and pedagogical significance. First, we extend the cross-linguistic evaluation of LLM-derived predictability and similarity measures from dominantly English settings to L2, lower-resourced, (Traditional) Chinese. Second, our interpretability analyses provide evidence of how these measures relate to the established linguistic dimensions and constructs and identify the conditions under which measures' interpretations may break down, thereby supporting more transparent and pedagogically meaningful applications of AI in writing assessment. Third, methodologically, we provide a reproducible and interpretable pipeline for systematically deriving LLM-based metrics, comparing them with classical linguistic metrics, and evaluating their incremental predictive value. We provide a structured approach for linking model-internal representations to linguistically interpretable constructs in L2 writing. By developing and testing this pipeline in Traditional Chinese (an underrepresented and comparatively lower-resource language in language model training data coverage and writing assessment), we demonstrate its

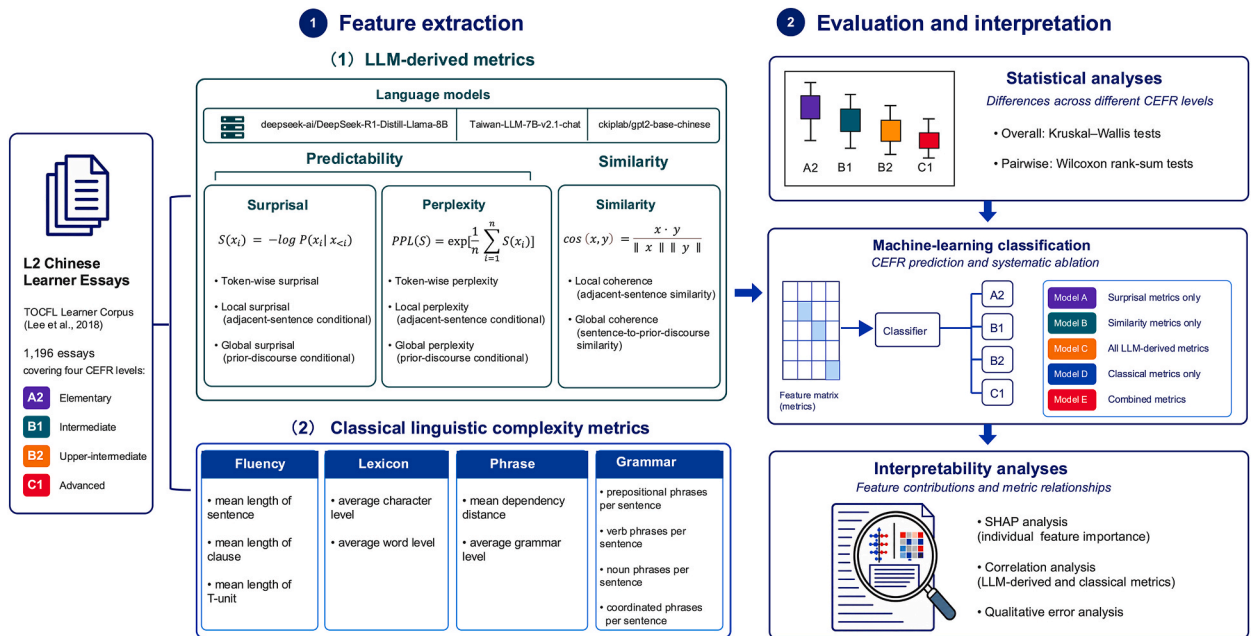


Fig. 1. Analytic pipeline integrating model-internal probability and embedding representations, classical linguistic metrics, proficiency classification, and interpretability analyses.

potential applicability to languages and learner populations for which task-specific computational resources remain limited. Finally, the cross-model comparisons suggest that model scale alone does not determine sensitivity to proficiency differences. Smaller, language-specific models can be as sensitive as—and, in some analyses, more sensitive than—larger general-purpose models when their training data and linguistic coverage are better aligned with the target language. Model–language alignment, rather than parameter count alone, is therefore an important consideration when selecting LLMs for transparent and resource-conscious L2 writing assessment in educational technology.

2. Background

2.1. Classical metrics for second language writing assessment

L2 writing is a multifaceted skill that requires learners to express ideas with accuracy, fluency, complexity, and coherence. While the CAF (complexity, accuracy and fluency) framework is central to research on L2 writing development and assessment (Housen et al., 2012; Polio, 2001; Vercellotti, 2017), coherence is often considered as a complementary dimension parallel to CAF (Verspoor & Rosmawati, 2024). Within this framework, Researchers have developed and validated a wide range of metrics to characterize L2 writing proficiency (e.g., Hao et al., 2024; Hu et al., 2022; Kyle & Crossley, 2015, 2018; Lu & Wu, 2022; Ortega, 2003). Lexical complexity includes lexical density, sophistication, and diversity, using indices such as type–token ratio, and frequency-based measures (e.g., the number of content words, function words or specific part of speech words). For instance, Zhang and Lei (2025) examined 50 lexical complexity metrics across these dimensions. Their multiple regression analysis found that higher-proficiency learners exhibited greater lexical diversity and sophistication in their writing. Phraseological complexity measures the structural patterns of multiword expressions and collocations, such as verb-object word combinations and adjective-noun word combinations (Hu et al., 2022). Syntactic complexity measures the diversity, sophistication, and elaboration of syntactic structures, including T-unit-based indices (Wu, 2016, 2018), topic-chain-based indices (e.g., ratio of different type of terminable topic-comment unite, Yu, 2021), dependency distance indices (e.g., Bi & Jiang, 2020), and phrasal sophistication (Lu, 2010). Hao et al. (2024) showed that both global and fine-grained syntactic complexity indices significantly predict L2 Chinese writing quality, with fine-grained measures stronger explanatory power. Fluency measures commonly used length-based indices (e.g., mean sentence length, Abdel Latif, 2013; Johnson et al., 2012; Plakans et al., 2019), and accuracy measures rely on error-based measures such as error-free units or error density (Liu & Shi, 2025). Coherence has been assessed using cohesion features (e.g., connectives), semantic overlap and semantic similarity measures such as latent semantic analysis and Word2Vec (Halliday & Hasan, 1976; McNamara et al., 2014). These indices provide interpretable but typically dimension-specific views of writing quality.

Many of these metrics can be automatically extracted using computational tools developed within natural language processing (NLP). Widely used toolkits such as Coh-Metrix (McNamara et al., 2014), L2 Syntactic Complexity Analyzer (Lu, 2010), TAASSC (Kyle, 2016), and TAACO (Crossley et al., 2016) enable large-scale analysis of learner writing by operationalizing these dimensions through a broad set of linguistic features. However, most established linguistic metrics and NLP tools were developed primarily for English (Malmasi & Dras, 2017; Xu et al., 2019), and comparable tool ecosystems remain far less developed for many other languages. This imbalance limits both the scale of automatic feature extraction and the cross-linguistic generalizability of feature-based assessment.

A complementary approach aligns learner production with standardized proficiency frameworks. For example, in L2 Chinese education, the *Chinese Proficiency Grading Standards for International Chinese Language Education* (Center for Language Education and Cooperation (CLEC), China's Ministry of Education, 2021; hereafter, the *Grading Standards*) provide an official proficiency framework that classifies Chinese characters, words, and grammatical structures into graded levels for language teaching and assessment. Assessment approaches based on the Grading Standards have been shown to effectively evaluate learners' writing proficiency (Wang et al., 2022).

Classical metrics discussed above are widely used in L2 writing assessment, but they generally capture individual dimensions of L2 writing. Automatic scoring models therefore combine large sets of hand-crafted features to approximate overall writing quality. For example, Wang and Hu (2021) used 90 linguistic features spanning multiple dimensions to build an automated essay scoring model to assess L2 Chinese writing. Although such approaches can achieve strong performance, they depend heavily on feature engineering, manual specification, and language-specific resources or NLP resources. These limitations motivate the need for more holistic and scalable metrics that can capture multiple dimensions of writing complexity in an integrated and interpretable manner. Moreover, discourse coherence is particularly difficult to quantify. Classical coherence metrics primarily rely on surface-level cohesion features such as lexical or grammatical cohesive devices (Crossley et al., 2016; Halliday & Hasan, 1976; Todirascu et al., 2013) and vector-based measures such as latent semantic analysis. These measures capture explicit connections between textual units but may not fully reflect the deeper semantic relations between sentences or paragraphs. These limitations motivate the need for more holistic and scalable metrics that remain interpretable and adaptable across languages.

2.2. Generative AI in second language learning and education

The rise of generative AI created new opportunities for second language learning and education. LLMs have shown potential in various educational applications, including automated essay scoring (Qiu et al., 2024; Stahl et al., 2024), grammatical errors detection (Mizumoto et al., 2024), conversational practice with chatbots (Xiao et al., 2023; Zheng et al., 2025), personalized and adaptive feedback (Liebenow et al., 2025), and generating customized learning or assessment materials (Kohnke et al., 2025; Pack & Maloney, 2024).

In L2 writing assessment, most applications of LLMs rely on prompt-based approaches, in which carefully crafted instructions are used to elicit scores, feedback, or error detection (Choi et al., 2026; Li & Liu, 2024; Saricaoglu & Bilki, 2025; Stahl et al., 2024). Evidence regarding their effectiveness, however, remains mixed. For instance, Mansour et al. (2024) found that LLMs underperformed compared to the state-of-the-art automated essay scoring systems, whereas Wang et al. (2025) showed that prompting enabled LLMs to conduct multidimensional assessments of L2 academic English writing and generate scores and comments according to predefined criteria. Evidence from broader higher education contexts further indicates that prompt configuration shapes assessment performance. AlGhamdi, Li, Gašević, and Chen (2025) compared prompting GPT-4o to perform scoring and feedback jointly in a single prompt or separately in consecutive prompts. Although differences in scoring performance were generally modest, errors were larger for lower-quality responses, and task order affected how the generated feedback supported learner agency. Collectively, these findings indicate that prompt-based assessment is sensitive to prompt structure, task sequencing, wording and decoding parameters such as model's temperature settings (Stahl et al., 2024; Tang et al., 2024). Such sensitivity can lead to inconsistent outputs and complicate the reproducibility and standardization required for educational assessment.

A more recent line of research has explored fine-tuning, whereby a pretrained model is further trained on task-specific data, such as scoring rubrics and human ratings. Fine-tuned models have generally outperformed their base models in automated essay scoring, achieving higher accuracy, reliability, and agreement with human ratings (e.g., Liu, Lu, & Qi, 2025; Wang & Gayed, 2026). For example, Liu, Qi, and Lu (2025) found that fine-tuned GPT models outperformed a range of linguistic complexity measures, while combining the two yielded only marginal improvements. Despite this, the models tended to overestimate essay quality and showed variability in performance across L1 groups. Nevertheless, the models tended to overestimate essay quality, and their performance varied across learners' L1 backgrounds. Fine-tuning also requires substantial labeled data, computational resources, and technical expertise, which may restrict its accessibility and scalability.

Thus, although prompt-based and fine-tuned approaches can generate effective assessment outcomes, both offer limited insight into how linguistic patterns represented *within* an LLM relate to writing quality. This limitation raises broader concerns about consistency, transparency, and construct interpretation. Rather than prompting or training an LLM to generate assessment scores directly, the present study examines model-derived probability and representation signals. Specifically, we investigate whether LLM-derived surprisal, perplexity, and embedding-based similarity can provide reproducible and linguistically interpretable indicators of variation in L2 Chinese writing quality and proficiency.

2.3. LLM-derived metrics for L2 writing assessment

Recent research has increasingly examined two types of model-derived signals for L2 writing assessment: probability estimates and contextual embeddings. Autoregressive language models are pretrained and optimized to predict the next token based on the preceding context using causal language modeling (Wolf et al., 2020) and can generate a conditional probability to each continuation given the preceding context. These conditional probabilities reflect how strongly the model anticipates each linguistic unit and can be transformed into measures of linguistic predictability such as surprisal. Embedding models instead represent tokens, sentences, or larger textual units as contextualized vectors that encode semantic information and relationships between textual units (Vaswani et al., 2017). These representations enable the calculation of semantic similarity. The following subsections review how probability-based predictability measures and embedding-based similarity measures have been used to characterize L2 writing and assess proficiency-related variation.

2.3.1. LLM-derived surprisal and perplexity as gradient measures of text naturalness

Surprisal is an information-theoretical measure that quantifies the unexpectedness of a linguistic unit given its preceding context under a language model (Hale, 2016; Misra, 2022). Lower surprisal value indicates that the observed continuation is more predictable or very likely to appear in context (Slaats & Martin, 2025). Perplexity expresses the same probability information on an exponential scale and reflects how well a language model predicts the observed text, with lower values likewise indicate greater predictability (Jurafsky & Martin, 2025). Although surprisal and perplexity are mathematically related, as both are derived from the same underlying probability information, they are expressed on different scales and are interpreted differently in language learning contexts. Conceptually, surprisal reflects the contextual unexpectedness of learner language to a language model, whereas perplexity reflects the model's average uncertainty in predicting the learner text. Lower values on both measures indicate that the text is more predictable under the language model.

Surprisal has been widely used in psycholinguistics to model human language processing difficulty based on the surprisal theory (Levy, 2008). Words with higher surprisal tend to elicit longer reading times and larger event-related potentials (N400 and P600) responses, reflecting the greater effort required to integrate less predictable input (Wilcox et al., 2023; Xu et al., 2023). Surprisal has recently been extended to the L2 writing assessment research. Cong (2025) developed an LLM-based pipeline for surprisal computation and found higher-proficiency L2 English writing exhibited lower surprisal, with lower surprisal being associated with greater fluency and phraseological complexity. Hu and Cong (2025) reported a comparable developmental pattern in L2 Chinese writing. Perplexity studies provide a related perspective. Sánchez et al. (2024) found that Swedish learner essays generally showed lower perplexity at higher proficiency levels and that normalized essays had lower perplexity than their original versions, although adjacent levels overlapped and some perplexity spikes reflected factors unrelated to learner competence. In longitudinal L2 French writing data, Godara et al. (2026) likewise identified average perplexity and surprisal as influential classification features and observed decreasing average surprisal over time.

These findings establish predictability measures including surprisal and perplexity as promising and holistic indicator of L2 writing

development, but several limitations remain. Evidence for perplexity has focused primarily on European languages, leaving its applicability to L2 Chinese unclear. Moreover, existing studies focus largely on token-level measures, while the value of sentence- and discourse-conditioned predictability remains underexplored. Fewer studies have tested whether surprisal and perplexity yield comparable proficiency-related patterns under controlled, matched contextual and modeling conditions. Also, the construct interpretation of surprisal remains uncertain. The present study addresses these gaps by comparing surprisal and perplexity across contextual scopes and models and by relating them to embedding-based similarity and classical linguistic metrics.

2.3.2. LLM-derived similarity as a semantic measure of discourse coherence

Discourse coherence is widely recognized as an important dimension of writing quality, yet it remains difficult to computationally operationalize. Traditional approaches have drawn on cohesion-based frameworks, notably Halliday and Hasan's (1976) classification of grammatical and lexical cohesion and have measured coherence through the frequency or ratio of explicit cohesive devices (i.e., Crossley et al., 2016; Peng et al., 2023). Although these cohesive devices assist in maintaining textual units and help distinguish coherent texts from disconnected sequences of sentences (Tanskanen, 2006), they primarily capture surface-level connections between textual units but may not fully represent the implicit semantic continuity that supports coherent discourse.

Vector representations offer a complementary approach. The textual embedding technique can represent words, sentences, paragraphs, or entire essays as numerical vectors that encode semantic information. Cosine similarity between these vectors provides a graded measure of semantic relatedness across textual units, allowing discourse coherence to be numerically operationalized as the degree of semantic continuity across a text. Earlier studies often used static embeddings such as Word2Vec or GloVe, which assign one vector representation to each word and therefore cannot capture context-dependent meaning. In contrast, contextualized language models generate representations that vary with the surrounding text, allowing the same lexical item to be represented differently across contexts. This distinction is particularly important for learner writing assessment, where meaning depends heavily on contextual use. Consistent with this advantage, Crossley and Holmes (2022) found that contextualized LLM embeddings outperformed static embeddings in predicting L2 learners' receptive vocabulary knowledge.

In L2 writing assessment, Cong (2024) used contextualized word and sentence embeddings to assess L2 English writing and found that higher-proficiency essays generally had lower similarity on several measures, although the pattern was not uniform across levels. Godara et al. (2026) similarly showed that semantic similarity features distinguished L1 and L2 French production and changed over time. These results motivate further work on how local and global similarity relate to discourse coherence and whether their developmental patterns generalize to languages beyond English and French.

2.4. The current study

Although prompted and fine-tuned LLMs can evaluate L2 learners' essays by generating essay scores and feedback, their sensitivity to task instructions and the limited transparency, stability, and reproducibility of their output constrain their use in high-stakes assessment. Explainable artificial intelligence (XAI) aims to make the behavior and outputs of AI systems understandable to relevant human stakeholders (Arrieta et al., 2020; Calderon & Reichart, 2025). Motivated by concerns about the opacity of black-box models (Rudin, 2019), we examine whether model-derived signals can serve as transparent and reproducible indicators of L2 writing quality. Accordingly, rather than treating directly generated scores and feedback as assessment outcomes, the present study uses two core LLM mechanisms (next token prediction and contextualized embeddings) to compute LLM-derived predictability and similarity metrics (Hale, 2001; Levy, 2008; Jurafsky & Martin, 2025) from learner texts. This approach supports methodological transparency and reproducibility while allowing the stability of these measures to be evaluated across contextual scopes and language models.

Building upon prior English L2 writing assessment studies that applied LLM-derived token-wise surprisal and similarity metrics (e.g., Cong, 2024; Sánchez et al., 2024), we validated these token-wise metrics and extend their measures to sentence and discourse contexts in L2 (Traditional) Chinese writing. Specifically, we computed local measures using the immediately preceding sentence as context and global measures using the full preceding discourse. These larger contextual windows capture how each sentence relates to its immediate and broader context, providing a richer account of predictability and semantic continuity in L2 writing. This multilevel approach is also motivated by the cross-linguistic difference in how discourse relations are expressed in English and Chinese: Unlike English often marks discourse relations through explicit lexical or grammatical connectives, Chinese coherence discourse may convey intersentential relations through semantic continuity, contextual inference, and topic chains rather than overt connectives (Halliday & Hasan, 1976; Wuyun, 2022; Yang, 2014). That is, English relies on grammatical cohesion, while Chinese prioritizes semantic cohesion (Chao, 1968; Li & Thompson, 1989). Consequently, L2 Chinese writing assessment would hypothetically benefit more from LLM-derived metrics that operate over *larger* contextual windows. Also, measures conditioned on adjacent sentences or preceding discourse may therefore capture information that is missed by token-level or surface-cohesion features.

The present study investigates the utility of LLM-derived probability metrics (surprisal and perplexity) and similarity metrics for L2 Chinese writing assessment, with a focus on their predictive power and interpretability. Specifically, we address three research questions.

- 1) Can LLM-derived surprisal, perplexity, and similarity metrics differentiate proficiency levels in L2 Chinese writing?
- 2) To what extent do LLM-derived metrics enhance the prediction of writing proficiency classification compared to classical linguistic metrics?
- 3) How are LLM-derived metrics related to classical metrics of lexical, phraseological, and syntactic complexity and fluency?

We hypothesize that both LLM-derived predictability and similarity are sensitive to L2 Chinese proficiency, but they capture different aspects of writing. Specifically, we expect higher-proficiency essays to show lower surprisal and perplexity, reflecting greater textual naturalness, and higher similarity, indicating more cohesive and semantically well-connected text. Because surprisal and perplexity are mathematically related probabilistic measures, they are expected to show similar proficiency-related patterns under matched conditions. Moreover, we conceptualize surprisal as a potentially integrative indicator of overall L2 writing proficiency, whereas embedding-based similarity is conceptualized as a more targeted indicator of semantic continuity—an important component of discourse coherence. Specifically, we expect surprisal to be associated with fluency and lexical, phraseological, and syntactic complexity, because model-based predictability may jointly reflect lexical choice, grammatical patterning, and contextual fit. We expect higher similarity to be associated with greater proficiency, reflecting stronger semantic connections across sentences and broader discourse. Because predictability and similarity capture related but distinct aspects of writing, we further expect LLM-derived metrics to provide additional predictive information when combined with classical linguistic metrics in writing proficiency classification.

3. Methods

3.1. Dataset

We used the publicly available Test of Chinese as a Foreign Language (TOCFL) Learner Corpus (Lee et al., 2018), which contains more than one million Chinese characters from Chinese learner essays. The TOCFL is a standardized language proficiency assessment test for non-native speakers of Chinese, and its writing component is aligned with Common European Framework of Reference (CEFR, Council of Europe. Council for Cultural Co-operation. Education Committee. Modern Languages Division, 2001). Essays were typed by examinees and evaluated by at least two trained professional Chinese teachers using a 0–5-point scale. Only essays with scores above 3 (the passing threshold) at each proficiency level were included in the TOCFL learner dataset. Therefore, this dataset has relatively high reliability and scoring consistency.

We extracted 1196 essays from the TOCFL Learner corpus across four CEFR proficiency levels (A2, B1, B2, and C1). Preprocessing removed extra spaces, newline characters, and non-characteristic symbols. Sentence boundaries were identified using Chinese punctuation (“。 ”!“ ”... ”“?”). Given that the surprisal calculations are affected by the text length, as the context window size can bias surprisal estimates (Hoover et al., 2023; Sánchez et al., 2024; Wilcox et al., 2023). Specifically, shorter texts may introduce greater variability and reduce comparability across samples, whereas longer texts tend to yield more stable surprisal estimates. Therefore, we followed Hu and Cong (2025) and truncated each essay to its first 250 Chinese characters when necessary. This threshold was motivated by the distribution of essay lengths in our dataset and retained a standardized and representative and comparable window to evaluate essay writing quality while reducing length-related variation. Table 1 summarizes essays count, genres, and lengths before and after the truncation.

3.2. LLM-derived metrics

All measures were computed programmatically to ensure reproducibility and efficient processing of large-scale text data. Both surprisal and sentences’ contextual embeddings were computed using Minicons (Misra, 2022), an open-source Python library that offers a unified interface for conducting behavioral and representational analyses of language models. Perplexity was calculated directly from model cross-entropy losses. As an overview, Table 2 summarizes the definitions, formulas, and computational procedures for all LLM-derived metrics.

(1) LLM-derived probabilistic metrics

Surprisal and perplexity are both derived from model probabilities but express predictability in different ways. Surprisal measures how unexpected a linguistic unit (e.g., a token, a word, or a sentence) is given its preceding context and is defined as the negative

Table 1
Descriptive statistics for essays count, genres, and lengths (measured in the number of characters).

CEFR Levels	Essay genres	# of essays	% in the dataset	Truncated text length		Original text length		
				Mean	Range	Mean	Range	# of truncated essays
A2 Elementary	practice notes; picture-description narratives	218	18.23	231.64	201–250	255	201–572	78
B1 Intermediate	personal letters; general narratives	380	31.77	249.27	209–250	391	209–799	368
B2 Upper-intermediate	functional writing; argumentative essays	502	41.97	249.75	153–250	558	153–1315	498
C1 Advanced	report writing; argumentative essays	96	8.02	223.28	147–250	522	147–1527	51

Table 2
Definitions and computation of the LLM-derived metrics.

Types	Formula	Metrics	Definitions
surprisal	For a target unit x_i , $S(x_i c) = -\log p(x_i c_i)$, where c_i denotes the conditioning context.	token-wise surprisal	surprisal of each token given its preceding context.
		local surprisal	surprisal of each sentence given the immediately preceding sentence
		global surprisal	surprisal of each sentence given the full preceding discourse.
perplexity	For an essay containing N tokens, perplexity was calculated as $PPL(x_{1:N}) = \exp \left[-\frac{1}{N} \sum_{i=1}^N \log P(x_i c_i) \right]$, where c_i denotes the conditioning context.	token-wise perplexity	Essay-level perplexity obtained by exponentiating the mean cross-entropy loss across all predicted tokens.
		local perplexity	Essay-level perplexity computed from all target-sentence tokens, with each target sentence conditioned on the immediately preceding sentence.
		global perplexity	Essay-level perplexity computed from all target-sentence tokens, with each target sentence conditioned on the full preceding discourse.
similarity	$\cos(e_a, e_b) = \frac{e_a \cdot e_b}{\ e_a\ \ e_b\ }$, where e_a, e_b are contextual embeddings of the textual units being compared.	local coherence (adjacent-sentence similarity)	cosine similarity between the embeddings of each sentence and the immediately preceding sentence
		global coherence (sentence-to-discourse similarity)	cosine similarity between each sentence embedding and the embedding of its full preceding discourse.

Note. Local measures (i.e., local surprisal/perplexity/coherence) are computed by conditioning on the immediately preceding sentence, whereas global measures are computed by conditioning on all preceding discourse. For each essay, token-wise surprisal was averaged across tokens, whereas local and global surprisal values were calculated from the second sentence onward and averaged across eligible sentences. Each perplexity measure yielded one essay-level score by exponentiating the mean negative log-likelihood over the eligible target tokens. Local coherence was averaged across adjacent sentence pairs, whereas global coherence was averaged across comparisons between each sentence and its full preceding discourse. Lower surprisal and perplexity indicate greater predictability, whereas higher coherence/similarity indicates stronger semantic continuity.

logarithm of its conditional probability (Hale, 2016; Misra, 2022). Lower surprisal indicates that the observed continuation is more strongly expected by the language model. Perplexity is the exponential of the mean negative log-probability assigned to a sequence of target tokens (Jurafsky & Martin, 2025) and summarizes the model's uncertainty when predicting those tokens. Lower perplexity values indicate that the text is easier for the model to predict.

Building on prior work with token-wise surprisal (Cong, 2025; Hu & Cong, 2025), we extended surprisal calculation to two sentence-level conditioning contexts. Adjacent-sentence conditional surprisal conditions each target sentence on the immediately preceding sentence (hereafter “*local surprisal*”). Discourse-conditioned surprisal instead conditions each target sentence on the full preceding discourse (hereafter “*global surprisal*”). As a supplementary analysis, we additionally computed token-wise perplexity (Sánchez et al., 2024), adjacent-sentence conditional perplexity (hereafter *local perplexity*), and perplexity conditioned on all previous contexts (hereafter “*global perplexity*”). Local and global calculations began with the second sentence and produced one essay-level score for each metric, as detailed in Table 2. This introduced perplexity alongside surprisal as a mathematically related but operationally distinct supplementary validation metric, allowing us to examine whether proficiency-related patterns were consistent across the two measures and whether sentence- and discourse-level contexts were more sensitive than token-wise measures.

We used autoregressive language models to calculate these metrics, as they were trained and optimized to predict the subsequent token in a sequence based on the preceding context using causal language modeling (Wolf et al., 2020), which directly corresponds to the definition of our probabilistic measures. In practice, we calculated surprisal and perplexity for each essay (written in traditional Chinese characters) with three autoregressive models: deepseek-ai/DeepSeek-R1-Distill-Llama-8B (Guo et al., 2025, hereafter “DeepSeek”), Taiwan-LLM-7B-v2.1-chat (Lin & Chen, 2023, hereafter “Taiwan-LLM”), and ckiplab/gpt2-base-chinese (CKIP Lab, 2021, hereafter “gpt2-base-chinese”). More specifically, DeepSeek is a distilled multilingual model with 8 billion parameters. Taiwan-LLM is a 7B parameter based on LLaMA-based model adapted with Traditional Chinese data; GPT2-base-chinese is a smaller model with 124 million parameters trained primarily on Traditional Chinese for general tasks. All selected LLMs are publicly available on HuggingFace (<https://huggingface.co/>, as of May 2025).

These models were selected because they differ in model scale, multilingual coverage, and alignment with the target language. DeepSeek provides broad multilingual coverage, whereas Taiwan-LLM and GPT2-base-chinese are more closely aligned with the Traditional Chinese essays examined in this study. We used the same models to calculate surprisal and perplexity for two purposes. First, it allowed us to examine whether the two probability-based measures produced comparable proficiency-related patterns. Second, it allowed us to evaluate whether these patterns were sensitive to model choice and whether language alignment might be as important as model size. Because raw probabilities values depend on tokenization and training distributions, we did not compare absolute surprisal or perplexity values across models; instead, we interpreted proficiency patterns within each model.

(2) Embedding-based similarity metrics

To assess semantic coherence in L2 writing, we calculated two LLM-derived similarity measures based on contextual embeddings: adjacent-sentence similarity and sentence-to-discourse similarity. The embeddings were generated the IDEA-CCNL/Erlangshen-

DeBERTa-v2-97M-Chinese model (Zhang et al., 2022), a transformer-based model pretrained on large-scale Chinese corpora, which can represent sentences and discourse contexts as context-sensitive vectors. Semantic coherence was operationalized as cosine similarity between contextualized sentence embedding vectors. Values closer to 1 indicate greater semantic relatedness between the two sentence representations and therefore strong coherence, whereas values closer to -1 indicate weaker semantic relatedness.

For each essay, we calculated adjacent-sentence similarity (hereafter, “*local coherence*”) and sentence-to-discourse similarity (hereafter, “*global coherence*”).

Local coherence was operationalized as the average cosine similarity between contextualized embeddings of adjacent sentences within an essay. Specifically, for each pair of immediately adjacent sentences, contextualized sentence embeddings were derived, and their cosine similarity was computed. The final local coherence score for each essay was obtained by averaging similarity values across all adjacent sentence pairs. Global coherence was operationalized as the average cosine similarity between each sentence and its preceding discourse context. Specifically, for each sentence i (starting from the second sentence), all preceding sentences (i.e., sentences 1 to $i-1$) were concatenated to form a discourse context. Contextualized embeddings were derived for both the current sentence and its preceding context, and cosine similarity was computed between the two representations. The final global coherence score for each essay was obtained by averaging these sentence-to-context similarity values.

Overall, LLM-derived similarity metrics capture local semantic continuity between adjacent sentences and broader consistency between individual sentences and the preceding discourse. We therefore hypothesized that higher cosine similarity would reflect greater semantic continuity or semantic coherence and would be associated with higher L2 writing proficiency.

3.3. Classical linguistic metrics

To provide established linguistic baselines for comparison with LLM-derived metrics, we included 11 classical linguistic complexity metrics widely used and validated in L2 writing assessment studies (Cui et al., 2022; Sung et al., 2016; Wang & Hu, 2021). These metrics capture multiple dimensions of writing, including lexical complexity, phraseological complexity, syntactic complexity, and fluency, as illustrated below.

- Lexical complexity was assessed using average character level and average word level. Both metrics are derived from a graded wordlist standard *Chinese Proficiency Grading Standards for International Chinese Language Education* (2021) and represent the average proficiency level of the characters and words used in an essay.
- Phraseological complexity was measured by the mean number of prepositional phrases, verb phrases, noun phrases, and coordinated phrases per sentence.
- Syntactic complexity was captured through mean dependency distance and average grammar level (also based on the *Grading Standards*).
- Fluency was measured by mean length of sentence, mean length of clause, and mean length of T-unit.

All these classical metrics were computed with two established NLP tools L2C-Rater (Wang & Hu, 2021) and CTAP for Chinese (Cui et al., 2022) and were included as baseline features in the subsequent machine learning and correlation analyses. Bonferroni-corrected pairwise Wilcoxon rank-sum tests showed that these classical metrics generally distinguished proficiency levels in the expected directions, supporting their robustness as baseline measures. Detailed results were reported in Appendix A.

3.4. Data analysis

To systematically evaluate LLM-derived metrics for L2 writing assessment, we employed a multi-stage data analysis framework integrating statistical testing, predictive modeling, and interpretability analyses, each corresponding to one of the study's research questions.

3.4.1. Proficiency-level analysis of LLM-derived metrics

We used Kruskal-Wallis tests to evaluate differences across the four CEFR levels for each surprisal, perplexity, and similarity measure. Significant omnibus tests were followed by Wilcoxon rank-sum tests for adjacent levels (A2 vs. B1, B1 vs. B2, and B2 vs. C1), with Bonferroni correction. Nonparametric tests were selected because visual inspection and Shapiro-Wilk tests indicated that the metric distributions did not satisfy the normality assumption. The alpha level was 0.05. Analyses were conducted in R (R version 4.5.1, R Core Team, 2025).

3.4.2. Machine learning analysis for writing proficiency classification

To test whether LLM-derived metrics provide incremental predictive value beyond classical linguistic measures in predicting learners' writing proficiency, we implemented the proficiency classification task under five feature configurations¹: (1) LLM-derived

¹ Perplexity was not included as a separate predictor because it was introduced as a supplementary validation measure and is derived from the same underlying probability information as surprisal. Also, as informed by Section 4.1 result, surprisal and perplexity exhibited highly similar patterns across proficiency levels. Including both would add redundant predictors and increase the risk of multicollinearity, we therefore retained surprisal as the primary probabilistic measure for the machine learning analyses.

surprisal metrics only, (2) LLM-derived similarity metrics only, (3) all LLM-derived surprisal and similarity metrics, (4) classical linguistic complexity metrics only, and (5) LLM-derived and classical metrics combined. This controlled ablation allowed us to quantify change in classification performance across different feature sets and pinpoint the predictive contribution of LLM-derived metrics. Four commonly used machine learning classifiers, including logistic regression, random forest, gradient boosting, and support vector machine (SVM), were used to evaluate under all five configurations to determine whether the results were consistent across modeling approaches. All machine learning classifiers were implemented in Python using scikit-learn (Varoquaux et al., 2015).

Following Cawley and Talbot (2010) and Cong (2024), we adopted the nested cross-validation approach to reduce the risk of model overfitting. The full dataset was randomly split into a training set (two-thirds of the data) and a held-out testing set (one-third of the data). All model training and hyperparameter tuning were conducted on the training set using nested 3-fold cross-validation, in which the inner loop performs a grid search for hyperparameter tuning, and the outer loop estimates generalization performance. The hyperparameter search space and model configuration settings are reported in Appendix B. Finally, after completing cross-validation, the model with the best-performing hyperparameter configuration was retrained on the full training set and evaluated on the held-out testing set. All classification performance reported were made on this unseen testing set. We considered proficiency prediction as a multi-class classification task. Model performance was assessed with accuracy (the proportion of correct predictions), precision (the proportion of correctly identified positive cases), recall (the model's ability to identify all relevant instances), and F1-score (the harmonic means of precision and recall), and the one-vs-rest AUC-ROC curve. To further examine how individual features contribute to model predictions, we applied SHAP (SHapley Additive exPlanations) analysis (Lundberg & Lee, 2017) to the best-performing model on the held-out testing set. By quantifying each feature's contribution, SHAP analysis helps unpack the “black box” of machine learning models and reveals which linguistic measures most strongly influence proficiency classification and improve model interpretability.

3.4.3. Interpretability analyses of LLM-derived metrics

We used Spearman rank correlations analysis to examine associations among LLM-derived surprisal, similarity, and classical metrics because the metrics were not non-normally distributed. The significance level was also set as $\alpha = .05$. We complemented the correlation analysis with a qualitative examination of essays whose surprisal patterns departed from the overall proficiency trend.

4. Results

4.1. Effectiveness of LLM-derived metrics in differentiating L2 writing proficiency

Before examining the predictive value of LLM-derived metrics in the classification task, we first investigated whether these metrics capture developmental differences in L2 Chinese writing. As shown in Table 3, the Kruskal–Wallis tests showed statistically significant differences across four proficiency levels for all 20 LLM-derived metrics (all $ps < 0.001$). The strongest effects were observed for metrics computed by GPT2-base-chinese: local perplexity ($\eta^2 = 0.294$), global perplexity ($\eta^2 = 0.279$), local surprisal ($\eta^2 = 0.244$), and global surprisal ($\eta^2 = 0.232$). Taiwan-LLM local and global surprisal also showed relatively large effects ($\eta^2 = 0.203$ and 0.197 , respectively). Across all three language models, local and global surprisal yielded larger effects than token-wise surprisal, indicating that sentence-level conditioning increased sensitivity to proficiency differences. This contextual advantage was less consistent for perplexity: local and global perplexity outperformed token-wise perplexity for GPT2-base-Chinese, whereas the pattern varied for Taiwan-LLM and DeepSeek. The similarity metrics also differentiated proficiency levels, with sentence-to-discourse similarity (global coherence, $\eta^2 = 0.164$) showed a larger effect than adjacent-sentence similarity (local global coherence, $\eta^2 = 0.096$).

Pairwise Wilcoxon rank-sum tests (with Bonferroni correction) further showed that the sensitive of LLM-derived probabilistic metrics depended on language model and conditioning context (see Fig. 2). Most LLM-derived surprisal and metrics statistically distinguished adjacent proficiency levels. For Taiwan-LLM, token-wise, local, and global surprisal decreased from A2 to C1 and distinguished every pair of adjacent levels. GPT2-base-Chinese local and global surprisal showed the same pattern, whereas token-wise surprisal did not distinguish A2 from B1. DeepSeek showed less consistent results: only local surprisal decreased monotonically and distinguished all adjacent levels. The results for Perplexity largely paralleled the surprisal findings, especially for Taiwan-LLM and GPT2-base-Chinese. DeepSeek perplexity, however, showed no consistent proficiency-related decline: token-wise and global perplexity increased from A2 to B2 before decreasing at C1, while local perplexity failed to distinguish the two lower-level contrasts. Thus, surprisal and perplexity produced closely aligned proficiency-related patterns for Taiwan-LLM and GPT2-base-Chinese, but not for DeepSeek. This provides a direct comparative analysis that confirms the robustness of our surprisal findings. Both similarity measures increased with proficiency (see Fig. 3). Global coherence (sentence-to-discourse similarity) distinguished all adjacent pairs, whereas local coherence (adjacent-sentence similarity) did not distinguish A2 from B1.

Overall, our results established that LLM-derived metrics were associated with proficiency level, but their sensitivity varied across models and conditioning contexts. Local and global measures generally provided clearer proficiency differentiation than token-wise measures, particularly for Taiwan-LLM and GPT2-base-Chinese, and global coherence was more sensitive than local coherence.

Another important implication is that the smaller gpt2-base-chinese model sometimes yields higher effect sizes compared to larger models. We maintain that while models like DeepSeek and Taiwan-LLM have significantly more parameters, GPT2-base-Chinese was

Table 3

Kruskal–Wallis tests of LLM-derived metrics across four proficiency levels, ordering by effect size.

LLM-derived metrics	χ^2 (3)	p value	η^2
local perplexity (GPT2-base-Chinese)	353.613	<0.001	0.294
global perplexity (GPT2-base-Chinese)	335.236	<0.001	0.279
local surprisal (GPT2-base-Chinese)	293.377	<0.001	0.244
global surprisal (GPT2-base-Chinese)	279.457	<0.001	0.232
local surprisal (Taiwan-LLM)	244.992	<0.001	0.203
global surprisal (Taiwan-LLM)	238.313	<0.001	0.197
local perplexity (Taiwan-LLM)	226.269	<0.001	0.187
global perplexity (DeepSeek)	214.535	<0.001	0.177
token-wise perplexity (DeepSeek)	207.036	<0.001	0.171
global coherence (sentence-to-discourse similarity)	197.923	<0.001	0.164
local surprisal (DeepSeek)	176.544	<0.001	0.146
token-wise perplexity (Taiwan-LLM)	131.279	<0.001	0.108
global perplexity (Taiwan-LLM)	131.898	<0.001	0.108
token-wise surprisal (Taiwan-LLM)	129.664	<0.001	0.106
adjacent-sentence similarity (local coherence)	117.986	<0.001	0.096
local perplexity (DeepSeek)	113.355	<0.001	0.093
global surprisal (DeepSeek)	105.093	<0.001	0.086
token-wise surprisal (GPT2-base-Chinese)	75.991	<0.001	0.061
token-wise perplexity (GPT2-base-Chinese)	59.802	<0.001	0.048
token-wise surprisal (DeepSeek)	52.332	<0.001	0.041

trained specifically on Traditional Chinese corpora. This narrower, more focused specialization may align more closely with the specific linguistic patterns and probabilistic expectations of the Traditional Chinese variety used in our learner corpus. In contrast, larger multilingual or distilled models may be over-regularized by broader data, making them slightly less sensitive to the specific developmental nuances of this learner population.

4.2. Predictive utility of LLM-derived metrics in L2 writing proficiency classification

We next examined whether LLM-derived measures improved automatic writing proficiency classification. To this end, we trained four classifiers, including logistic regression, support vector machine (SVM), random forest, and gradient boosting, to evaluate multiclass classification performance using five feature sets using controlled ablation design: LLM-derived surprisal metrics only (model A), LLM-derived similarity metrics only (model B), LLM-derived metrics (model C), classical metrics only (model D), and both LLM-derived and classical metrics (model E). This allows the contribution of surprisal, similarity, and classical features to be compared under a common evaluation protocol.

Table 4 summarizes the classification performance (accuracy, precision, recall, and F1 score) for each machine learning classifier. Overall, adding LLM-derived metrics to classical metrics (model D vs. model E) improved proficiency classification performance across classifiers. SVM achieved the best performance overall: its F1-score increased from 0.678 with classical metrics alone to 0.719 with the combined feature set. These gains indicate that LLM-derived metrics added predictive information not fully captured by the classical linguistic metrics and complement these established measures in automatic L2 writing proficiency classification.

We then used SHAP analysis to examine individual feature contributions in the best-performing model (the SVM). As shown in Fig. 4, average lexical level and average character level were the most influential predictors for writing proficiency classification. Among the LLM-derived measures, Taiwan-LLM global, token-wise, and local surprisal also ranked highly, while local and global coherence made moderate contributions. GPT2-base-Chinese token-wise surprisal and several syntactic features (e.g., mean length of sentence) made smaller contributions. The classifier therefore relied primarily on lexical sophistication and Taiwan-LLM-derived surprisal, with similarity providing complementary information.

To evaluate level-specific discrimination, we also trained one-vs-rest random-forest classifiers. Each CEFR level (A2, B1, B2, C1) was treated as the positive class against all remaining levels (binary classification). We report the mean and standard deviation of AUC, with values closer to 1 indicating stronger discrimination. Given the imbalanced dataset, we used a stratified two-fold cross-validation to preserve the class distribution in each fold. Model performance was evaluated using the area under the receiver operating characteristic curve (AUC), and we report the mean and standard deviation across the two folds. Values closer to 1 indicate. The model, including both LLM-based and classical metrics (Model E), produced the highest AUC for all four CEFR levels and was particularly effective for distinguishing A2 and C1 (Fig. 5). This pattern was consistent with the multiclass classification results reported in Table 4 and further supports the incremental predictive value of LLM-derived metrics.

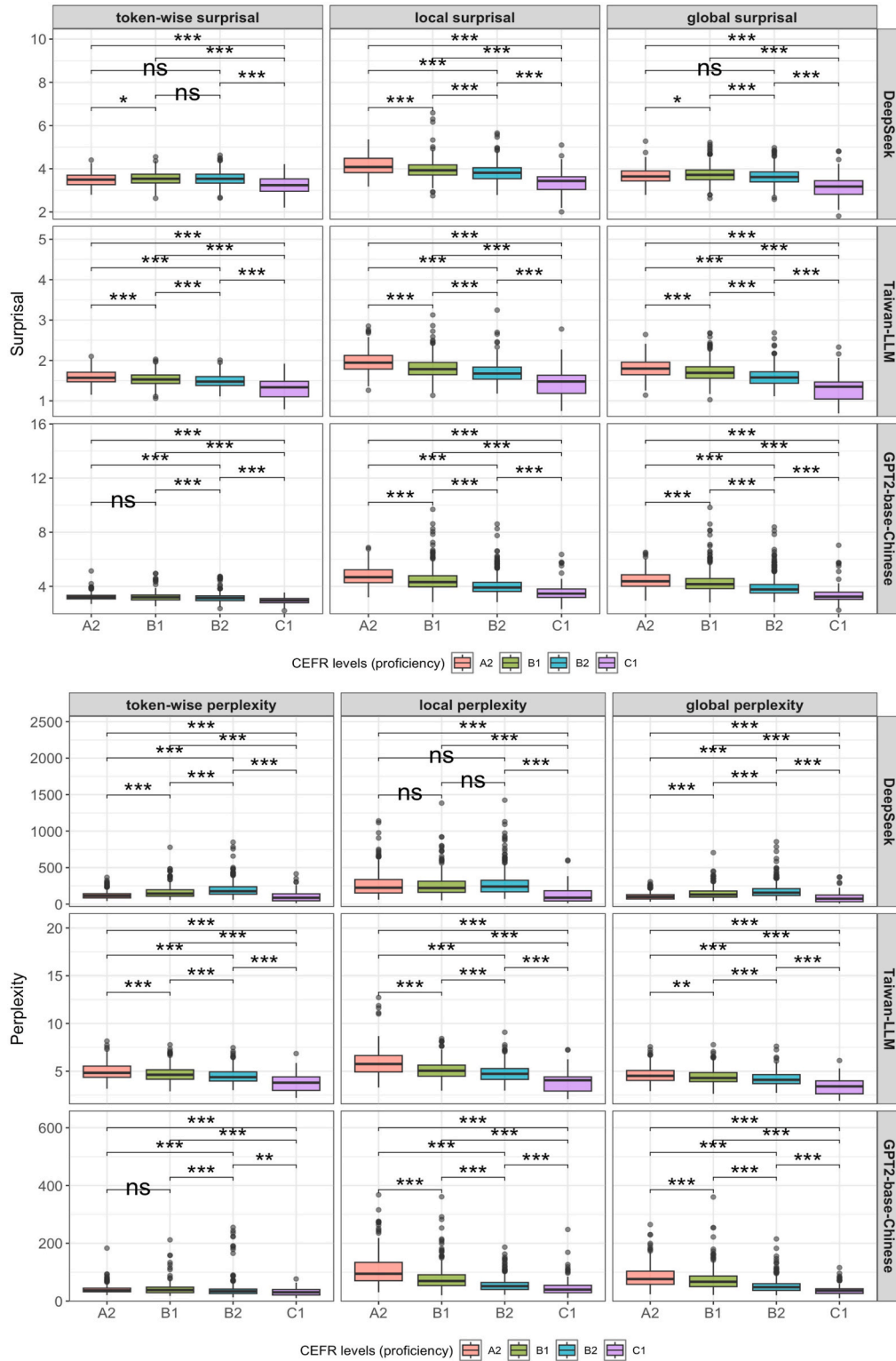


Fig. 2. LLM-derived surprisal and perplexity across CEFR proficiency levels. Pairwise comparisons between proficiency levels were conducted using Wilcoxon rank-sum tests with Bonferroni correction. Asterisks indicate statistically significant differences (Significance notation: * $p < .05$, ** $p < .01$, *** $p < .001$; ns = not significant).

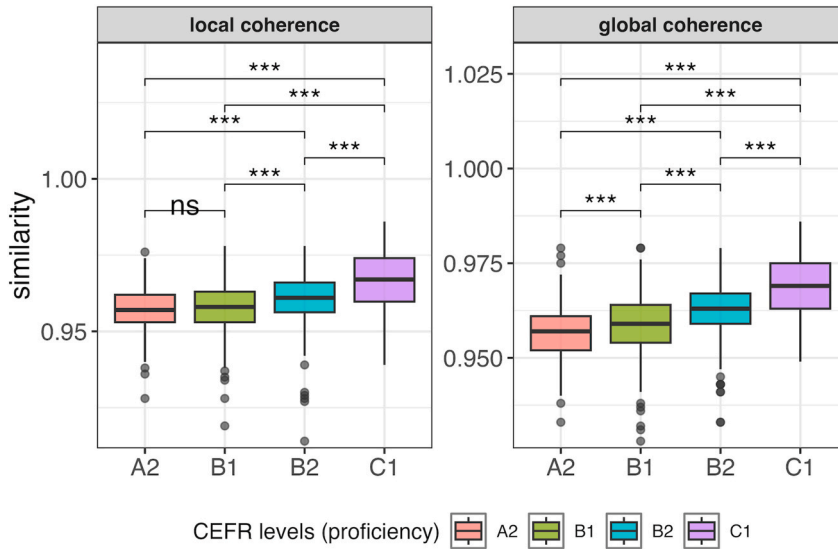


Fig. 3. Local and global coherence metrics across CEFR levels (Significance notation: *** $p < .001$; ns = not significant).

4.3. Relationships between LLM-derived metrics and classical linguistic metrics

Fig. 6 shows that local and global surprisal were more strongly related to classical linguistic metrics than token-wise surprisal. For Taiwan-LLM and GPT2-base-Chinese, local and global surprisal were significantly negatively correlated with length-based fluency metrics (i.e., mean length of sentence, clause, and T-unit, $r_s = -0.70$ to -0.33), average character and word levels ($r_s = -0.35$ to -0.41) and mean dependency distance ($r_s = -0.49$ to -0.36). Correlations with average grammar level were minimal ($r_s = -0.12$ to -0.07). Together with the proficiency-level results in Section 4.1, these associations indicate that essays produced by more proficient learners tended to show lower contextual surprisal alongside more advanced character and vocabulary use, longer linguistic units, and greater dependency-based syntactic complexity. Thus, surprisal was associated with multiple dimensions of L2 writing, although its relationship with grammar-level complexity was limited.

Both similarity measures positively correlated with length-based fluency metrics ($r_s = 0.33$ – 0.68), average character and word levels ($r_s = 0.25$ – 0.34), mean dependency distance ($r_s = 0.42$ – 0.50), and phraseological metrics ($r_s = 0.22$ – 0.40). Global coherence showed stronger and more consistent correlations with these classical metrics than local coherence. Together with the proficiency-level results in Section 4.1, these correlations indicate that essays produced by more proficient learners tended to exhibit stronger semantic continuity alongside longer linguistic units, more sophisticated lexical choices, and greater dependency-based and phraseological complexity. Additionally, similarity metrics were correlated moderately and negatively with surprisal metrics, particularly those derived from Taiwan-LLM. That is, essays with higher semantic coherence tended to be more predictable. Notably, while surprisal and coherence metrics are significantly correlated, we maintain that they are not overlapping. Specifically, surprisal acts as a holistic index of writing naturalness, whereas embedding-based similarity specifically targets semantic continuity and discourse coherence.

Since correlation analysis captures the general association pattern between LLM-derived metrics and classical metrics, we additionally examined unexpected cases through the qualitative error analysis (see Table 5). The A2 and B1 texts are both narrative both narratives, yet GPT2-base-Chinese assigned lower surprisal to the A2 essay despite the B1 essay's greater lexical and syntactic diversity. One possible explanation lies in differences in topic predictability and discourse structure. The A2 essay focuses on highly conventional topics such as season and weather, which are associated with highly predictable expressions. In contrast, the B1 essay develops a temporally ordered narrative, where each event builds the previous one. While this linear progression creates coherence, it also expands the range of plausible continuations at each step, making the text less predictable for the language model and resulting greater surprisal. Moreover, a clearer, unexpected increase in surprisal emerges when comparing B1 and B2, where the genre shifts from narrative to argumentative writing. Unlike narratives, which typically follow a temporally ordered sequence of events, argumentative texts require the development of abstract reasoning and the organization of ideas. This shift is accompanied by changes in lexical choice, with B2 texts containing more abstract and less frequent vocabulary. As a result, we maintain that the predictability of upcoming content is reduced, leading to unexpectedly higher surprisal values. However, even when controlling the same argumentative genre (B2 vs. C1), the expected decrease in surprisal with higher proficiency is not observed. Instead, the C1 essay shows higher

Table 4

Performance of four machine learning classifiers across five different feature configurations for predicting Chinese learners' essay proficiency (A2, B1, B2, C1).

ML models	classification performance	LLM-derived surprisal metrics (model A)	LLM-derived similarity metrics (model B)	all LLM-derived (metrics (model C)	classical metrics (model D)	all LLM-derived and classical metrics (model E)
Logistics Regression	accuracy	0.561	0.495	0.551	0.693	0.713
	precision	0.566	0.498	0.541	0.696	0.715
	recall	0.561	0.495	0.551	0.693	0.713
	F1-score	0.537	0.451	0.530	0.690	0.711
Gradient Boosting	accuracy	0.548	0.464	0.543	0.652	0.685
	precision	0.538	0.444	0.539	0.659	0.688
	recall	0.548	0.464	0.543	0.652	0.685
	F1-score	0.539	0.436	0.535	0.650	0.685
SVM	accuracy	0.538	0.487	0.586	0.683	0.718
	precision	0.551	0.471	0.614	0.688	0.732
	recall	0.538	0.487	0.586	0.683	0.718
	F1-score	0.521	0.439	0.567	0.678	0.719
Random Forest	accuracy	0.541	0.429	0.563	0.637	0.680
	precision	0.535	0.419	0.560	0.639	0.693
	recall	0.541	0.429	0.563	0.637	0.680
	F1-score	0.529	0.421	0.556	0.631	0.676

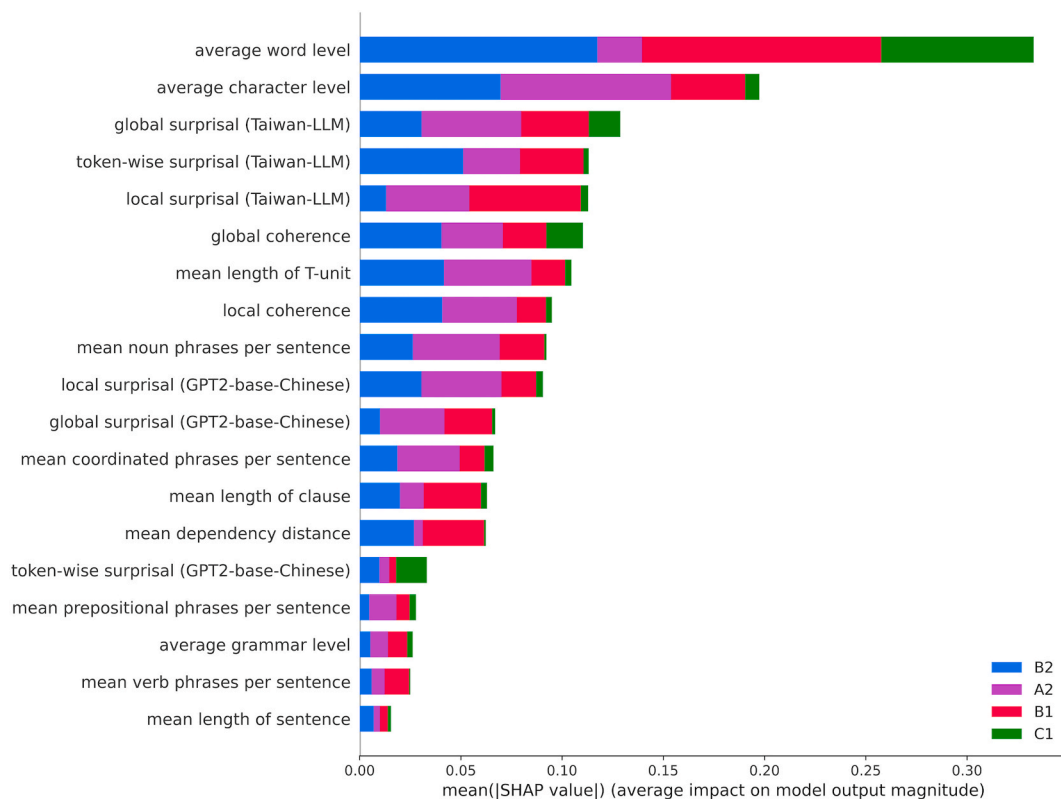


Fig. 4. Feature importance on SVM predicting Chinese learners' essay proficiency level (A2, B1, B2, C1), ranked based on the absolute SHAP value.

surprisal than B1. While the B2 text maintains a relatively linear and constrained argumentative structure, the C1 essay exhibits topic shifts (e.g., from pet rental to abstract reflections on value), philosophical statements (e.g., “動物如同人物...” *animals are like humans*), and rhetorical questions (e.g., “真正的價值有多少?” *how worthy is it anyways?*), all of which expand the space of possible continuations. This reduced discourse constraint makes the C1 text less predictable for the language model, resulting in increased surprisal. The lack of explicit discourse markers (e.g., “因此” *therefore*, “總之” *overall*) weakens the coherence between segments, making it difficult for the model to maintain stable expectations. Notably, this pattern is partially reflected in similarity metrics. While local coherence remains comparable between the two texts (B2: 0.958; C1: 0.957), global coherence shows divergence (B2: 0.954; C1: 0.962), suggesting that similarity metrics capture certain aspects of discourse organization but may not fully align with surprisal in cases of abstract or weakly constrained discourse. Taken together, these cases point to a key limitation of surprisal: its sensitivity to predictability is not always aligned with proficiency when discourse-level constraints are weak. In particular, texts that involve abstract reasoning, open-ended questioning, or loosely connected ideas introduce a broader range of plausible continuations, leading to elevated surprisal values that do not directly reflect writing quality. By contrast, similarity appears to be less affected by such variation, as it captures semantic continuity and discourse-level relatedness across sentences. The divergence observed here suggests that surprisal and similarity operate over different dimensions of text, and that their combination can provide a more balanced account of writing quality.

5. Discussion

This study examined whether LLM-derived probabilistic and similarity metrics can function as robust and interpretable indicators of L2 writing development. Local and global surprisal generally differentiated proficiency more clearly than token-wise surprisal, and perplexity reproduced this pattern for the Traditional Chinese-specialized models but not consistently for DeepSeek. Similarity metrics can also capture proficiency-related differences in semantic continuity/coherence, particularly when each sentence was compared with its preceding discourse. Combining LLM-derived and classical linguistic features consistently improved writing proficiency classification.

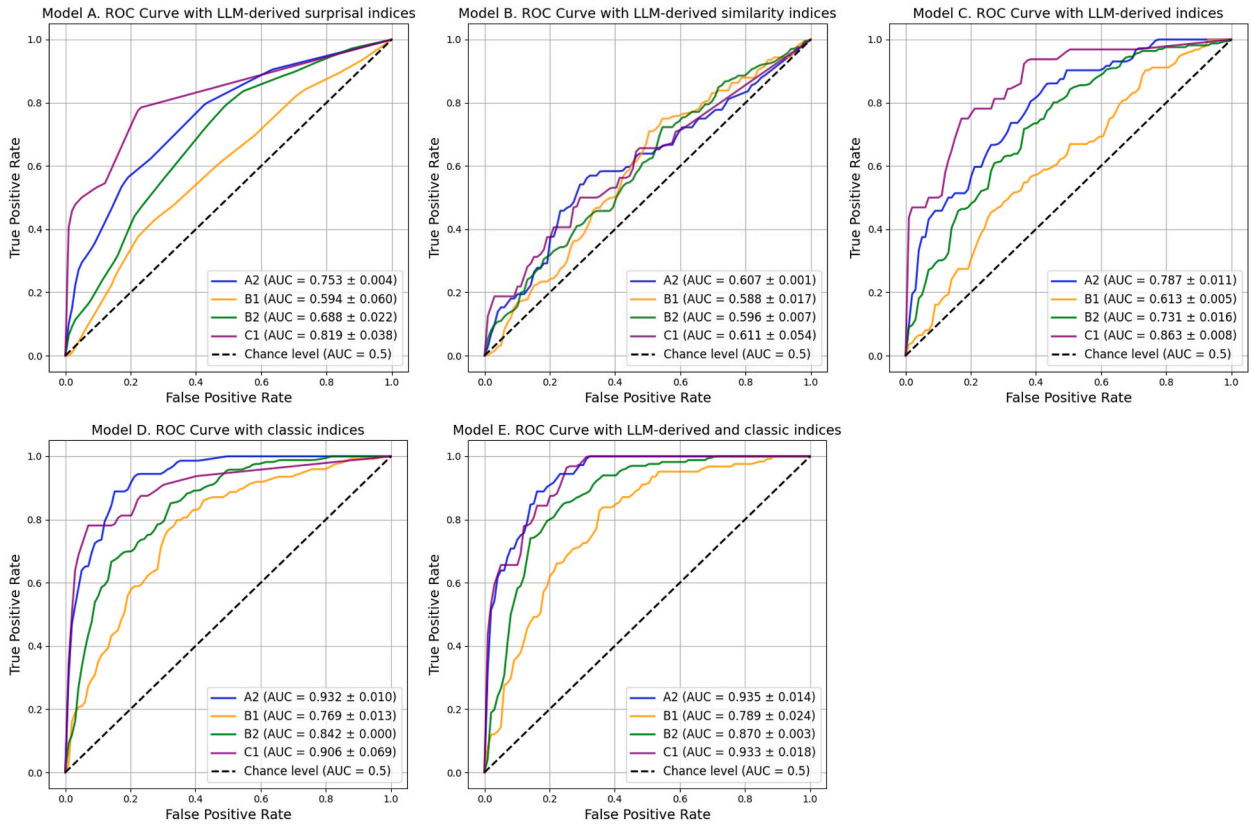


Fig. 5. Comparison of multiclass ROC curves and AUC values for One-vs-Rest random forest classifiers across five feature sets: (A) LLM-derived surprisal metrics only, (B) LLM-derived similarity metrics only, (C) all LLM-derived metrics, (D) classical metrics only, and (E) combined LLM-derived and classical metrics.

5.1. Interpreting LLM-derived metrics as indicators of L2 writing development

LLM-derived surprisal reflects the extent to which linguistic input conforms to a language model's probabilistic expectations and can therefore be interpreted as an index of writing naturalness. Our finding that higher-proficiency essays received lower surprisal scores from Taiwan-LLM and GPT2-base-Chinese is consistent with previous research on L2 English (Cong, 2025), supporting surprisal's cross-linguistic applicability for L2 writing development research. Whereas prior studies focused primarily on token-level predictability, the present study further showed that local and global surprisal were more sensitive to proficiency differences and more strongly related to classical linguistic measures. In L2 Chinese, the lower surprisal observed in higher-proficiency essays co-occurred with more advanced character and vocabulary use, longer linguistic units, and greater phraseological and syntactical complexity. Its relationships with similarity metrics further suggest that surprisal may therefore function as a potentially integrative indicator of writing quality, as surprisal is influenced not only by lexical usage and grammatical well-formedness, but also by how well the discourse follows culturally and linguistically expected patterns.

The perplexity analysis's results largely reproduced the surprisal patterns for Taiwan-LLM and GPT2-base-Chinese. For both models, perplexity generally decreased as proficiency increased, consistent with previous DeepSeek perplexity did not show the expected decline. This contrast suggests that model-language alignment may be particularly important for underrepresented or comparatively lower-resource languages. Smaller, language-specific models may be more sensitive to proficiency differences when their training data and tokenization better match the target language.

We demonstrate that, with non-English datasets and models, LLM-derived similarity provides a semantic proxy for discourse coherence by quantifying semantic continuity or relatedness between sentences and their preceding contexts. We found that both local and global coherence increased with proficiency, with global coherence showing clearer proficiency differentiation and stronger relationships with classical linguistic metrics. This pattern contrasts with Cong's (2024) findings for L2 English, where higher proficiency was associated with lower similarity on several measures. The difference may be both methodological and linguistic.

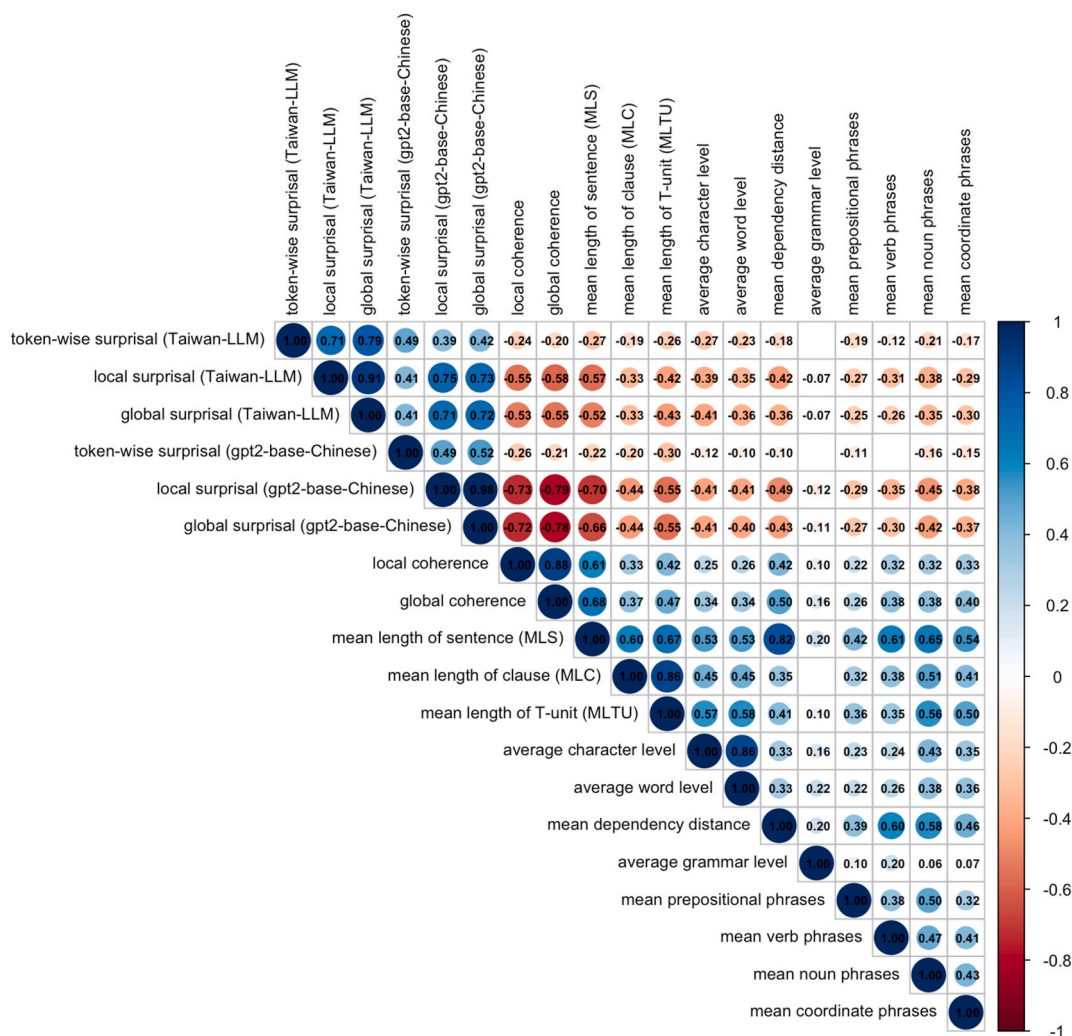


Fig. 6. Spearman correlation coefficients heatmap of LLM-based metrics and classical metrics (Darker colors indicate stronger correlations. Insignificant cells are left blank). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Methodologically, our global coherence measure compares each sentence with its full preceding discourse, thereby emphasizing broader thematic continuity rather than local lexical repetition. Linguistically, Chinese discourse often relies on topic continuity and contextual inference to connect ideas (Malmasi & Dras, 2017; Peng et al., 2023; Xu et al., 2019), which may increase the sensitivity of discourse-level similarity measures. Therefore, we speculate that higher global coherence observed among more proficient learners may reflect a developing ability to maintain topics and organize meaning across extended discourse.

5.2. Implications for L2 teaching, learning, and assessment practice

From an L2 teaching and learning perspective, LLM-derived surprisal and coherence metrics offer fine-grained, reproducible indicators for tracking L2 writing development, as continuous measures can reveal gradual changes that may be obscured by categorical proficiency labels. This is potentially useful for advanced L2 learners' writing, where development may be incremental and distributed across lexical, syntactic, and discourse dimensions. Because the present study used cross-sectional proficiency data, however, longitudinal research is needed to establish whether these metrics can track individual learners' development over time.

Table 5

Essay excerpts with unexpectedly high or low GPT2-base-Chinese surprisal and their similarity score. Full texts are provided in [Appendix C](#). Certain text were truncated for privacy reasons.

levels	genre & topic	essay samples	GPT2-base-Chinese surprisal & similarity
A2	narrative my hometown	我住在xx可以看到每個季節。夏天天氣很熱,冬天很冷可是下雪非常美。秋天的天氣也還不錯,也很漂亮,因為可以看到很多的開花。 I live in xx and can enjoy every season. The weather is very hot in summer and very cold in winter, but the snow is very beautiful. The autumn weather is also nice and beautiful because we can see many flowers blooming.	token-wise surprisal = 2.697 local surprisal = 3.849 global surprisal = 3.675 local coherence = 0.958 global coherence = 0.966
B1	narrative my favorite day	我妈妈来的那天,我非常期待,上午上课时,我没像平常专心地听课,人虽然在教室里但心只想时间过快一点而让我看到妈妈。下了课,我没吃午餐就直接搭客运往xx机场去。 The day my mother arrived, I was extremely excited. During the morning class, I didn't listen attentively as I usually do. Although I was in the classroom, my mind was only focused on how much time would pass so that I could see my mother. After class, I didn't have lunch and immediately took the bus to xx Airport.	token-wise surprisal = 3.237 local surprisal = 4.001 global surprisal = 0.959 local coherence = 0.958 global coherence = 0.962
B2	argumentative My views on pet rentals	我非常支持這樣寵物出租的方式。因為許多一個人住的人,不管男女老少,都想要養寵物,但是實際不容易實現的事。... ...有的給別人,有的留在公園裡等。 I strongly support this method of renting out pets. Because many people, regardless of gender or age, who live alone wish to have pets, but it is actually difficult to achieve in reality. ... Some are given to others, while others are left in the park.	token-wise surprisal = 3.343 local surprisal = 4.204 global surprisal = 0.958 local coherence = 0.959 global coherence = 0.960
C1	argumentative My views on pet rentals	首先,我反對這種新興行業,因為,無論小狗可愛或熱情,它還是會被視為「東西」... 動物如同人物:它們也有感覺,而且也知道愛是什麼。... 人想要有狗陪伴而覺得太麻煩,自私了! ... 可是一定不會有客人去想,小狗給他的快樂,真正的價值有多少?所以,那些小狗將來會有特別的利嗎?我確定應該不會吧! First of all, I oppose this emerging industry because, regardless of how cute or enthusiastic the puppy is, it will still be regarded as "an object" ... Animals are like people: they have feelings and they also know what love is. ... People think it's too troublesome to have a dog as a companion, and they are being selfish! ... But surely no one would think about how much true value the happiness brought by the puppy actually holds. So, will those puppies have any special advantages in the future? I'm pretty sure they won't!	token-wise surprisal = 3.547 local surprisal = 5.742 global surprisal = 5.662 local coherence = 0.957 global coherence = 0.955

Importantly, these metrics may also support diagnostic approaches to writing instruction and intelligent tutoring systems (Martynova et al., 2025) which provide individualized feedback and adaptive learning paths. At the passage level, unusually high surprisal can draw attention to lexical choices or constructions that are unexpected in context, while low local or global coherence can indicate possible disruptions in topic maintenance or semantic continuity. Such information could help teachers and learners identify passages for closer review and develop more targeted feedback on lexical and grammatical choices, textual naturalness, and discourse organization. However, these signals should not be treated automatically as evidence of error or poor writing. High surprisal may also reflect creativity, abstraction, or a legitimate topic shift, while lower coherence may result from the introduction of necessary new information. Accordingly, teachers might consider interpreting these signals in relation to the genre, task purpose, and broader discourse context rather than treat them as stand-alone indicators of writing quality. With respect to assessment practice, this study shows that LLM-derived metrics can complement existing classical linguistic indices and human judgment within automated writing assessment pipelines. Unlike prompted holistic scores, these metrics are derived directly from language model-internal probabilistic and representational mechanisms, enhancing transparency, replicability, and interpretability and addressing practical concerns in AI-enhanced assessment. When combined with classical metrics, these LLM-derived metrics support more efficient and informative assessment while retaining sensitivity to multiple dimensions of L2 proficiency. Importantly, the observed differences across language models underscore the need for careful model selection in applied contexts, as alignment between model training data and target learner language is crucial for assessment. These LLM-derived metrics support more efficient and informative assessment while retaining sensitivity to multiple dimensions of L2 proficiency. Importantly, the observed differences across language models underscore the need for careful model selection in applied contexts, as alignment between model training data and target learner language is crucial for assessment.

These implications also resonate with recent discussions in educational NLP, which emphasize the importance of aligning LLM capabilities with the needs of educational stakeholders and pedagogical principles (Pal Chowdhury et al., 2025). From the perspective of teacher training and professional development, LLM-derived metrics can support educators in interpreting and critically engaging with AI-generated assessments, rather than adopting them as opaque scoring tools. Furthermore, LLM-derived metrics are not replacements for teachers or traditional assessment frameworks, but as components of hybrid human-AI approaches that integrate theoretical insights from L2 writing research with emerging technological affordances.

5.3. Limitations and future directions

Several limitations should be acknowledged. First, the corpus was imbalanced across CEFR levels and limited to examination writing in Traditional Chinese. In addition, variation in prompts and genres may partly account for differences in predictability, as illustrated by qualitative cases. Although the observed patterns were consistent across various analyses and cross-validation was

utilized to reduce the risk of overfitting, subsequent research should investigate whether LLM-derived surprisal and similarity metrics demonstrate comparable effectiveness across larger, more balanced datasets and diverse genres to evaluate their generalizability.

Second, the present study used text truncation as a compromise to control variation in essay length. This improved comparability but did not preserve each learner's complete response. Future research should compare full-text analyses with standardized windows and examine whether the relative performance of local and global measures changes with essay length or collect responses under a common length constraint and the same topic.

Third, the findings highlight the model-dependent nature of LLM-derived metrics. Future research should therefore systematically compare a broader range of LLMs, including models trained on different writing systems and with different objectives (e.g., base vs. instruction-tuned models), to better understand how model characteristics influence LLM-derived metric behaviors.

Fourth, the classical metrics covered complexity and fluency but not writing accuracy. In future work, we plan to incorporate fine-grained grammatical error annotations, covering both word-level categories (e.g., part-of-speech usage) and higher-level syntactic functions. This will allow us to systematically investigate how different types of learner errors (e.g., incorrect lexical selection, word order, or functional misuse) relate to LLM-derived surprisal values. Such analysis will further clarify whether surprisal is sensitive to accuracy-related deviations or primarily reflects distributional predictability independent of grammatical correctness.

Finally, while this study focused on the interpretability and developmental relevance of LLM-derived metrics, we did not directly examine how these metrics relate to human numerical ratings, instructional decisions, or learning outcomes. An important direction for future research is to connect these computational indices with teacher judgments, learner reflections, and pedagogical interventions in authentic instructional contexts. Such work would help situate LLM-derived metrics within classroom practice and assessment of workflows, strengthening the link between theoretical modeling, empirical analysis, and educational application. Despite these limitations, our work provides methods and pipelines that could shed light on educational technology in an under-resourced language learning context.

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CRediT authorship contribution statement

Jingying Hu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Yan Cong:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of interest statement

The authors declare no conflict of interest.

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Appendix A. classical metrics' effectiveness in differentiating L2 writing proficiency

Table A1

Kruskal-Wallis test results: the efficacy of classical metrics in differentiating writing proficiency levels

classical metrics	χ^2 (3)	<i>p</i> value	η^2
mean length of sentence (MLS)	399.649	<0.001	0.333
mean length of clause (MLC)	292.006	<0.001	0.242
mean length of T-unit (MLTU)	454.894	<0.001	0.379
average character level (char_avg_level)	627.421	<0.001	0.524
average word level (lex_avg_level)	635.192	<0.001	0.53
mean dependency distance (mean_dep_dist)	228.823	<0.001	0.189
average grammar level (gram_avg_level)	39.791	<0.001	0.031
mean number of preposition phrases per sentence	120.580	<0.001	0.099
mean number of verb phrases per sentence	133.321	<0.001	0.109
mean number of noun phrases per sentence	310.745	<0.001	0.258
mean number of coordinate phrases per sentence	231.027	<0.001	0.191

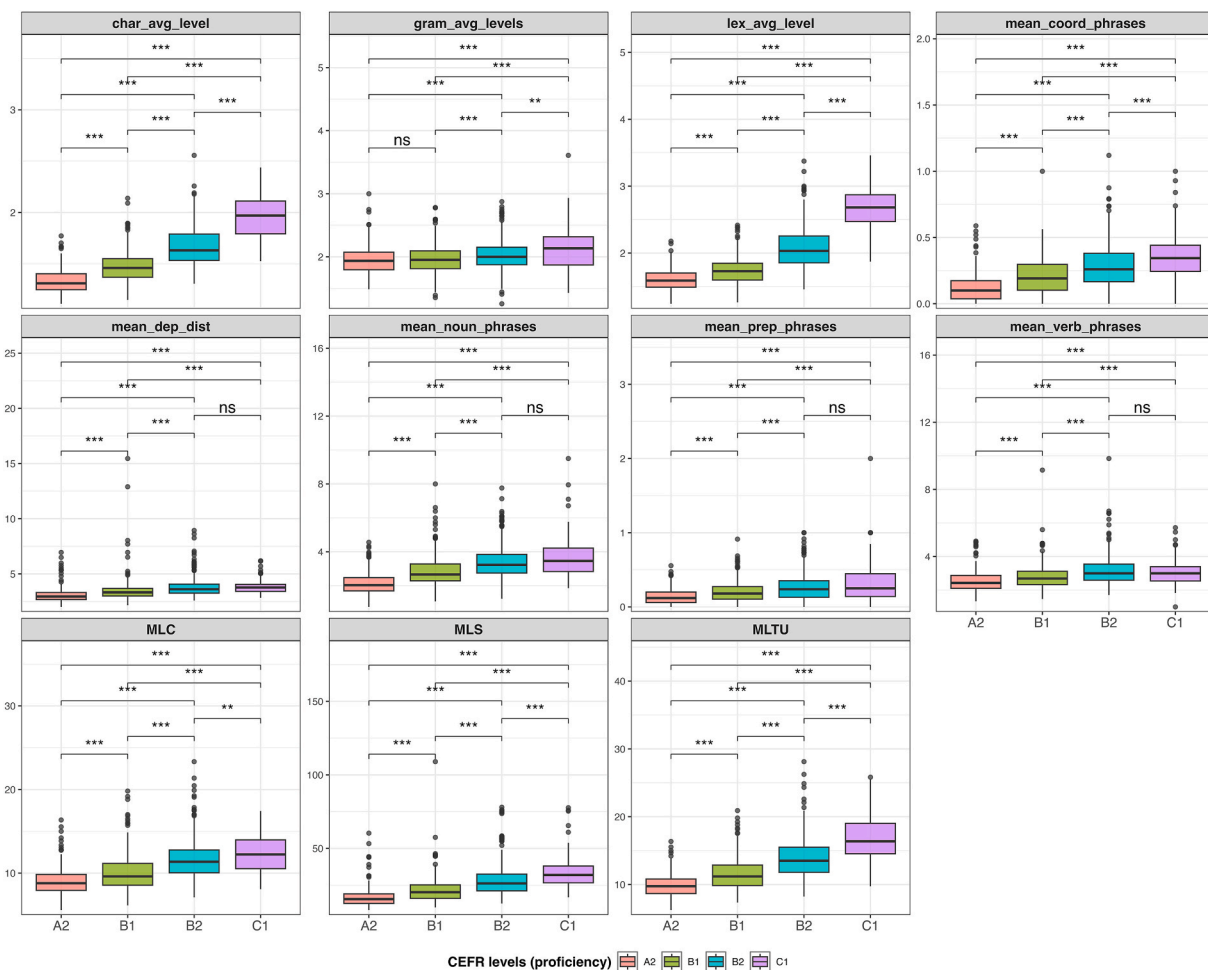


Fig. A2. Paired comparisons across learner proficiency levels for selected classic metrics

Appendix B. machine learning (ML) classifiers' hyperparameter tuning

Table B1

Machine learning (ML) classifiers hyperparameter tuning: grid search space

ML models	Grid search space
Logistics Regression	Regularization parameter (C): 0.01, 0.1, 1, 10, 100 penalty: 'l2'
Gradient boost	The function to measure the quality of a split (criterion): [friedman_mse, squared_error] The number of boosting stages to perform (n_estimators): [100,200,300,400]
SVM	The number of features to consider when looking for the best split (max_features): ['log2', None] ('sqrt' means using $\sqrt{2}$ (total number of features), 'log2' means using $\log_2$ (total number of features)) Regularization parameter (C): [0.1, 1, 10, 100] Kernel coefficient (gamma): [1, 0.1, 0.01, 0.001, 0.0001]
Random Forest	Kernel type to be used in the algorithm (kernel): [rbf, sigmoid] The function to measure the quality of a split (criterion): [gini, entropy, log_loss] The number of trees (n_estimators): [100,200,300,400] The number of features to consider when looking for the best split (max_features): ['sqrt', 'log2', None] ('sqrt' means using $\sqrt{2}$ (total number of features), 'log2' means using $\log_2$ (total number of features))

Table B2

Machine learning (ML) classifiers hyperparameter selected after tuning

feature combination	ML models	Hyperparameters selected after tuning
only LLM-derived surprisal metrics (model A)	Logistics Regression	C:10, penalty: l2
	Gradient boost	criterion: squared_error, max_features: sqrt, n_estimators: 100
	SVM	C: 10, gamma: 0.1, kernel: rbf
	Random Forest	criterion: entropy, max_features: None, n_estimators: 10
only LLM-derived similarity metrics (model B)	Logistics Regression	C:100, penalty: l2
	Gradient boost	criterion: friedman_mse, max_features: sqrt, n_estimators: 100
	SVM	C: 10, gamma: 1, kernel: rbf
	Random Forest	criterion: entropy, max_features: sqrt, n_estimators: 300
LLM-derived surprisal and similarity metrics (model C)	Logistics Regression	C:100, penalty: l2
	Gradient boost	criterion: squared_error, max_features: log2, n_estimators: 100
	SVM	C: 100, gamma: 0.01, kernel: rbf
	Random Forest	criterion: entropy, max_features: None, n_estimators: 400
Only classical linguistic metrics (model D)	Logistics Regression	C:1, penalty: l2
	Gradient boost	criterion: friedman_mse, max_features: sqrt, n_estimators: 400
	SVM	C: 10, gamma: 0.01, kernel: sigmoid
	Random Forest	criterion: entropy, max_features: sqrt, n_estimators: 100
LLM-derived and classical linguistics metrics (model E)	Logistics Regression	C:1, penalty: l2
	Gradient boost	criterion: friedman_mse, max_features: sqrt, n_estimators: 200
	SVM	C: 100, gamma: 0.001, kernel: sigmoid
	Random Forest	criterion: entropy, max_features: sqrt, n_estimators: 400

Appendix C. qualitative error analysis essay samples

Table C1

Concrete examples from learners with different proficiency levels and the corresponding LLM-derived surprisal and similarity values. Certain text were truncated for privacy reasons.

levels	genre & topic	essay samples	GPT2-base-Chinese surprisal & similarity
A2	記敘文 我的故鄉	我在美國長大的。我住xx住了差不多二十一年。xx的天氣有的時候很好,有的時候不怎麼好。我住在xx可以看到每個季節。夏天天氣很熱,冬天很冷可是下雪非常美。秋天的天氣也還不錯,也很漂亮,因為可以看到很多的開花。春天的天氣比較好,我覺得很舒服,我最喜歡這個季節,因為春天的時候什麼地方都可以看起來很美,我最喜歡春天的樹葉的顏色,我最喜歡的地方在xx是美國的最大的動物中心,我在這裡沒什麼都可以買到,可是有很多的東西我買不起。	token-wise surprisal = 2.697 local surprisal = 3.849 global surprisal = 3.675 local coherence = 0.958 global coherence = 0.966
B1	記敘文 我最快樂的一天	我刚来xx时非常想家,虽然已经xxx岁,曾经离开家庭去很远的地方暂时工作,但那种出国留学,三个月没跟家人见面的感觉真的让人伤心。我父母的感觉也没什么两样。所以我妈妈来xx时,一看到她,我就开心得手舞足蹈起来。那是我二十四年生活当中最快乐的一天。我妈妈十月底来xx,一方面因为很想念我而要求来看我,一方面跟她的朋友一起旅行。我妈妈来的那天,我非常期待,上午上课时,我没像平常专心地听课,人虽然在教室里但心只想时间过快一点而让我看到妈妈。下了课,我没吃午餐就直接搭客运往xx机场去。	token-wise surprisal = 3.237 local surprisal = 4.001 global surprisal = 0.959 local coherence = 0.958 global coherence = 0.962
B2	議論文 我對寵物出租的看法	我非常支持這樣寵物出租的方式。因為許多一個人住的人,不管男女老少都想要養寵物,但是實際不容易實現的事。原因很多,比如說有租房的規定,因工作太忙而早出晚回的習慣,不能照顧等等。但是有些人卻忍不住養寵物,可是過了幾個月就放棄飼養。有的給別人,有的留在公園裡等,完成沒有責任性。這樣情況使許多人很難過。還有造成流浪寵物不會減少的情形。那寵物出租公司的最優點是出租以前,有面試並訓練課程,這很有信用的事。生意是生意,但是也有責任性。先了解寵物的養法是實在、重要的。寵物受到很多愛情,飼主也受到幸福感。	token-wise surprisal = 3.343 local surprisal = 4.204 global surprisal = 0.958 local coherence = 0.959 global coherence = 0.960
C1	議論文 我對寵物出租的看法	首先,我反對這種新興行業,因為,無論小狗可愛或熱情,它還是會被視為「東西」,而因此人不會去想此動物的狀況如何,更不會去想其未來的生活。動物如同人物:它們也有感覺,而且也知道愛是什麼。我十分服膺這句話,因為我知道那些看準了這商機的人,這樣做只是為了錢而已。基本上,若人太忙,那麼根本不應養寵物。人想要有狗陪伴而覺得太麻煩,自私了!對人來說,付一筆租金,就可以把小狗租回家玩幾天,太簡單,可是一定不會有客人去想,小狗給他的快樂,真正的價值有多少?所以,那些小狗將來會有特別的利嗎?我確定應該不會吧!	token-wise surprisal = 3.547 local surprisal = 5.742 global surprisal = 5.662 local coherence = 0.957 global coherence = 0.955

Data availability

All analysis code used in this study are publicly available on the Open Science Framework (OSF) at: <https://doi.org/10.17605/OSF.IO/QFK2H>.

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